

Monetary policy: Inflation targeting in an open economy pt. 3

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1 Inflation targering in an SOE

Transmission mechanism

- The open-economy IS-curve:

$$y = -\alpha_1(r - \rho) + \alpha_2 e + v$$

where $e = s + p^* - p$ is the real exchange rate (in logs)

- The open-economy Phillips curve:

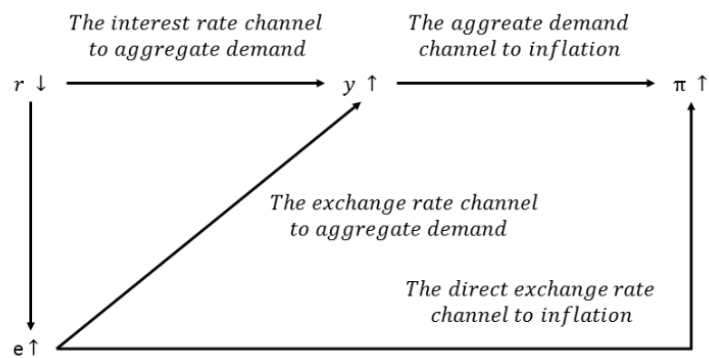
$$\pi = \pi^e + \gamma_1 y + \gamma_2 + u$$

- The real UIP:

$$e = e^e - (r - r^*) + z \iff r = r^* + (e^e - e) + z$$

The monetary policy transmission mechanism

A reduction in the key policy rate



Monetary policy

- The central bank minimizes the loss

$$L = \frac{1}{2} [(\pi - \pi^*)^2 + \lambda y^2]$$

- The optimality condition is

$$\pi - \pi^* = -\frac{\lambda}{\gamma_1 + \frac{\gamma_2}{\alpha_1 + \alpha_2}} y$$

which again can be written as

$$y = -\frac{\gamma_1 + \frac{\gamma_2}{\alpha_1 + \alpha_2}}{\lambda} (\pi - \pi^*)$$

- Reduced-form IS equation:

$$y = -(\alpha_1 + \alpha_2)(r - \bar{r})$$

where $\bar{r} = \rho + \frac{1}{\alpha_1 + \alpha_2} v + \frac{\alpha_2}{\alpha_1 + \alpha_2} ((r^* - \rho) + e^e + z)$

- Reduced-form Phillips curve:

$$\pi = \pi^2 + \left(\gamma_1 + \frac{\gamma_2}{\alpha_1 + \alpha_2} \right) y + u^{\text{open}}$$

where $u^{\text{open}} = u + \frac{\gamma_2}{\alpha_1 + \alpha_2} (\alpha_1(r^* - \rho + e^e + z) - v)$

- The optimality condition is

$$\pi - \pi^* = -\frac{\lambda}{\gamma_1 + \frac{\gamma_2}{\alpha_1 + \alpha_2}} y$$

1.1 Graphical analysis

- To simplify the figures, we let $\pi^e = \pi^*$, $r^* = \rho$ and $e^e = 0$

1. The (y, π) -diagram

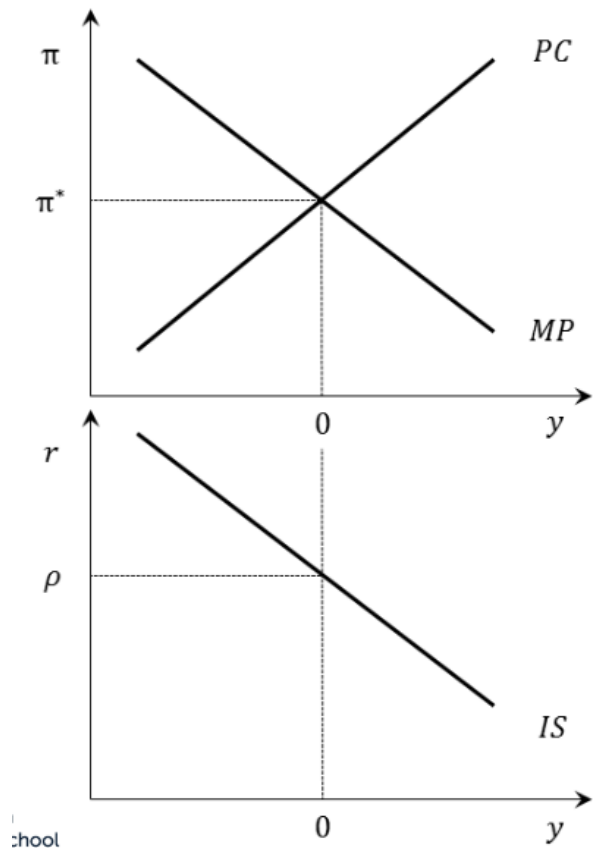
- We use the reduced-form PC and thereby we take into account that inflation depends on the real exchange rate
- MP — with slope $-\frac{\lambda}{\gamma_1 + \frac{\gamma_2}{\alpha_1 + \alpha_2}}$

2. The (y, r) -diagram

- IS given by:

$$r = \bar{r} - \frac{1}{\alpha_1 + \alpha_2} y$$

The MP-PC-IS chart



Negative demand shock

